No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

No. of Functional Features Included in the Solution

Project: OptiCrop — Smart Agricultural Production Optimization Engine

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	**Crop Recommendation**	Accepts N, P, K, temperature, humidity, pH, and rainfall inputs and uses the trained Random Forest model to recommend the most suitable crop for those exact conditions.	ML Prediction Layer	**Done**	5
2	**User Input Form**	A clean web form on the Find Your Crop page that collects all seven soil and climate parameters from the farmer with clearly labelled input fields.	Frontend / UI	**Done**	5
3	**Agricultural Data Processing**	Processes raw user inputs by converting them to float values, assembling them into a NumPy array, and scaling them using the saved StandardScaler before passing them to the model.	Backend / Flask	**Done**	5
4	**Multi-Algorithm Model Comparison**	Trains and evaluates four supervised classifiers — Logistic Regression, KNN, Decision Tree, and Random Forest — on identical train/test splits and selects the best-performing model.	ML Pipeline / Notebook	**Done**	5
5	**K-Means Clustering Analysis**	Applies K-Means clustering with the Elbow Method (k=4) to discover natural agro-climatic groupings in the dataset, supporting research and policy insights.	EDA / Notebook	**Done**	5

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
6	**Responsive Web Interface**	Three-page Flask application (Home, About, Find Your Crop) styled with Bootstrap 5 for a clean, mobile-compatible layout usable by any farmer without ML expertise.	Frontend / UI	**Done**	5
7	**Input Validation & Error Handling**	Validates all seven form fields on submission; non-numeric or missing inputs are caught by a try/except block and return a plain error message without crashing the server.	Backend / Flask	**Done**	5
8	**GitHub Repository & Version Control**	Complete project hosted at github.com/neelimavana/OptiCrop with meaningful commit history, structured folder layout, and a detailed README for reproducibility and mentor review.	Repository / Git	**Done**	5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Crop Recommendation, User Input Form, Data Processing, Recommendation Engine, Input Validation, Error Handling)
Additional / Nice-to-Have Features	2 (K-Means Clustering Analysis, GitHub Repository & Version Control)
Features Tested & Verified	8
Overall Marks Contribution	40 Marks (8 features × 5 marks each)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	**User Interface (UI)**	User Input Form, Responsive Interface	Find Your Crop page form with 7 labelled input fields (N, P, K, Temperature, Humidity, pH, Rainfall). Bootstrap 5 navigation across Home, About, and Find Your Crop pages.
2	**Backend / Logic**	Data Processing, Recommendation Engine, Input Validation, Error Handling	Flask routes handling GET/POST requests. StandardScaler transforms inputs. RandomForestClassifier.predict() generates crop label. try/except block catches invalid inputs gracefully.
3	**ML Pipeline / Analysis**	Multi-Algorithm Comparison, K-Means Clustering	Training and comparing Logistic Regression, KNN, Decision Tree, and Random Forest. Elbow Method for K-Means (k=4). Confusion matrix and classification report for Random Forest (99.55% accuracy).

S.No	Category	Features in Category	Example Features
4	**Model Persistence / Serving**	Crop Recommendation (model serving)	model.pkl (Random Forest), scaler.pkl (StandardScaler), label_encoder.pkl (LabelEncoder) loaded once at Flask app startup and reused for every prediction request.
5	**Version Control / Repository**	GitHub Repository	Full project at github.com/neelimavana/OptiCrop — includes source code, dataset, trained model artifacts, Jupyter notebook, requirements.txt, and README with setup instructions.